Post-hoc Interpretability of LDA on Hyperspectral Imagery: SHAP Attributions, Counterfactual Topic Flips, and LLM-judge Alignment under Token-Mass-Dispersion Asymmetry

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***Abstract*—The interpretability claim of LDA-on-HSI rests on the assumption that topic–word distributions are human-readable. *This study is restricted to the LDA backbone*; the cross-backbone comparison (HDP, ProdLDA, ETM) is the subject of the companion paper P4. We test three orthogonal post-hoc interpretability axes on the V1–V15, V17–V20 wordification sweep (eighteen LDA-fitted recipes) from the companion paper P3: F-13 SHAP attribution of pixel-to-topic decisions via a closed-form posterior, F-22 counterfactual L_1 perturbation required to flip a document's argmax topic, and F-15 LLM-as-judge alignment between a document's top tokens and its argmax topic's top tokens. We find three concrete results. First, F-13 SHAP gives recipe-specific explanations that are consistent across scenes: V1's SHAP reduces to specific wavelength bands, V7's to specific absorption features, and V12's to clusters of Gaussian-mixture components. Second, F-22 counterfactual L1 (sentinel-patched 19-recipe ranking) separates the recipes into "robust topic" (V20 = 26.3, V12 = 24.5, V3 = 23.5), "moderate" (V1 = 6.1, V7 = 5.2) and "fragile" (V9 = 1.0, V10 = 1.2) regimes; the most-robust basis (V20, the label-aware MI-weighted recipe) is $\sim$26$\times$ harder to flip than V9's. Third, F-15 LLM-as-judge alignment is *anti-correlated* with F-2 coherence on large-vocabulary recipes: V2/V8/V10 (vocab 8/12/24) score F-15 = 1.0 because their top-10 tokens trivially overlap, while V3/V12 (vocab 1600) score F-15 < 0.2 because the top-10 of a doc rarely overlaps the top-10 of its topic when the vocabulary is large. The confounder is *not purely vocabulary size*, however: V20 re-weights V3's identical $(\mathrm{band}, \mathrm{bin})$ alphabet and matches V12's same-sized GMM-token alphabet (mean nominal $|\mathcal{V}| \approx 1376$ across the six scenes, $B\,Q$ per scene), yet scores F-15 = 0.64 against their 0.12 / 0.16 — so cardinality cannot explain the gap. We make the real mechanism quantitative with the *token-mass dispersion* metric, the effective vocabulary $N_{\text{eff}} = \exp\big(H(\phi_k)\big)$ of a topic's word distribution (averaged over topics and scenes): N_{eff} falls V3 $(431) >$ V12 $(400) >$ V20 (309) and F-15 *rises* in lock-step $(0.12 < 0.16 < 0.64)$, whereas a pure-cardinality account predicts no difference. The artefact is therefore driven by token-mass dispersion, not nominal vocabulary cardinality. This exposes a methodological gap: F-15 as currently defined conflates token-mass concentration with coherence, and an LLM oracle would face the same artefact unless given semantic — not identity — comparison capability. We recommend reporting F-13 SHAP top-K attributions as the primary interpretability artefact, F-22 as the topic-stability artefact, and F-15 only after dispersion normalisation.**

***Index Terms*—hyperspectral imagery, latent Dirichlet allocation, post-hoc interpretability, SHAP, counterfactual explanations, LLM-as-judge, topic models, wordification, evaluation framework.**

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$$\phi_k \sim \mathrm{Dir}(\beta), \quad \theta_d \sim \mathrm{Dir}(\alpha), \quad z_{dn} \sim \mathrm{Cat}(\theta_d), \quad w_{dn} \sim \mathrm{Cat}(\phi_{z_{dn}}). \tag{1}$$

$$\boldsymbol{\beta} \sim \mathrm{GEM}(\gamma), \quad \pi_d \sim \mathrm{DP}(\alpha_0, \boldsymbol{\beta}), \quad z_{dn} \sim \mathrm{Cat}(\pi_d), \quad w_{dn} \sim \mathrm{Cat}(\phi_{z_{dn}}) \tag{2}$$

with global atom weights $\beta_k = \beta'_k \prod_{l<k}(1-\beta'_l)$, $\beta'_k \sim \mathrm{Beta}(1, \gamma)$.

$$\delta_d \sim \mathcal{N}(\mu_0, \Sigma_0), \quad \theta_d = \mathrm{softmax}(\delta_d), \quad w_{dn} \sim \mathrm{Cat}\big(\mathrm{softmax}(\boldsymbol{\beta}\,\theta_d)\big), \tag{3}$$

where $\boldsymbol{\beta} \in \mathbb{R}^{|V| \times K}$ is *unnormalised* (the word distribution is a product of experts, not a per-topic simplex), and (μ_0, Σ_0) is the Laplace approximation to $\mathrm{Dir}(\alpha)$; inference is amortised, $(\mu_d, \Sigma_d) = \mathrm{Enc}_\eta(n_d)$.

$$\beta_k = \mathrm{softmax}\big(\rho^\top \alpha_k\big), \qquad \theta_d = \mathrm{softmax}(\delta_d), \quad w_{dn} \sim \mathrm{Cat}\Big(\textstyle\sum_k \theta_{dk}\,\beta_k\Big) \tag{4}$$

with word embeddings $\rho \in \mathbb{R}^{L \times |V|}$ and topic embeddings $\alpha_k \in \mathbb{R}^L$. (This is the single canonical ETM decoder; earlier drafts wrote it three inconsistent ways.)

$$\tilde{x}_{db} = \frac{x_{db} - \min_b x_{db}}{\max_b x_{db} - \min_b x_{db}}, \qquad q_b(x_d) = \mathrm{clip}\big(\lfloor \tilde{x}_{db}\, Q \rfloor,\, 0,\, Q{-}1\big). \tag{5}$$

$$n_d^{\mathrm{V1}}[b] = q_b(x_d), \qquad |V_{\mathrm{V1}}| = B. \tag{6}$$

$$n_d^{\mathrm{V3}}[(b, q)] = \mathbb{1}[q_b(x_d) = q], \qquad |V_{\mathrm{V3}}| = B\,Q. \tag{7}$$

$$a_d = \arg\min_{a \geq 0} \|x_d - E\,a\|_2^2, \qquad n_d^{\mathrm{V8}}[m] = \big\lfloor s\, a_{dm} \big\rceil, \tag{8}$$

where $E = [e_1, \ldots, e_M]$ are the NFINDR endmembers (simplex vertices) and s a fixed integer scale; $|V_{\mathrm{V8}}| = M$.

$$r_{dk} = \frac{\pi_k \mathcal{N}(x_d \mid \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_d \mid \mu_j, \Sigma_j)}, \qquad n_d^{\text{V12}}[(k,q)] = \mathbb{1}\big[q_b(r_{d\cdot}) = q\big]. \tag{9}$$

$$w_b = \Big\lfloor \frac{\text{MI}(x_{\cdot b}; y)}{\max_{b'} \text{MI}(x_{\cdot b'}; y)} \, Q_{\max} \Big\rceil, \qquad n_d^{\text{V20}}[(b,q)] = w_b \mathbb{1}[q_b(x_d) = q]. \tag{10}$$

The token alphabet is V3's, $|V_{\text{V20}}| = B\,Q$ (=1600 at $B = 200, Q = 8$; mean ≈ 1376 across the six labelled scenes since B varies). Bands with $w_b = 0$ emit nothing, so the *effective* (non-empty-column) vocabulary is smaller still (≈ 770 mean).

$$\text{F-1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2\,P_c R_c}{P_c + R_c}. \tag{11}$$

$$\text{NPMI}(w_i, w_j) = \frac{\log \frac{p(w_i, w_j) + \epsilon}{p(w_i) p(w_j)}}{-\log\big(p(w_i, w_j) + \epsilon\big)}, \quad c_v = \frac{1}{N} \sum_i \cos\big(\mathbf{v}_i, \textstyle\sum_j \mathbf{v}_j\big), \tag{12}$$

with $\mathbf{v}_i = \big(\text{NPMI}(w_i, w_\ell)\big)_{\ell=1}^{N}$ over the top-N topic words.

$$\hat{z}_d = \arg\max_k \theta_{dk}, \qquad \text{F-7} = \frac{I(\hat{Z}; Y)}{H(Y)}. \tag{13}$$

$$\text{F-14} = \frac{2}{K(K-1)} \sum_{k<k'} \frac{|T_k \cap T_{k'}|}{|T_k \cup T_{k'}|}, \qquad T_k = \text{top-}N(\phi_k). \tag{14}$$

$$\text{F-18} = \frac{1}{\binom{S}{2}} \sum_{s<s'} \frac{1}{K} \max_{\sigma \in \mathfrak{S}_K} \sum_k \cos\big(\mathbf{u}_k^{(s)}, \mathbf{u}_{\sigma(k)}^{(s')}\big), \tag{15}$$

over S reseeded refits, with $\mathbf{u}_k^{(s)}$ the top-N indicator vector of topic k in run s.

$$\text{F-22}(d) = \min_{\delta \in \mathbb{Z}^{|V|}} \|\delta\|_1 \ \text{ s.t. } \ \arg\max_k \theta_k(n_d + \delta) \neq \arg\max_k \theta_k(n_d), \ n_d + \delta \geq 0, \tag{16}$$

reported as the per-scene median, capped at a sentinel $\text{MAX\_STEPS} = 50$ when no flip is found.

$$\text{aligned}(d) = \mathbb{1}\Big[\,|\,\text{top}_N(n_d) \cap \text{top}_N(\phi_{\hat{z}_d})\,| \geq \tau\,\Big], \quad \text{F-15} = \tfrac{1}{D} \textstyle\sum_d \text{aligned}(d). \tag{17}$$

$$p_{dv} = \frac{n_{dv}}{\sum_{v'} n_{dv'}}, \qquad N_{\text{eff}}(d) = \exp\big(H(p_d)\big) = \exp\Big(-\sum_v p_{dv} \log p_{dv}\Big) \tag{18}$$

Low N_{eff} (mass concentrated on few tokens) keeps top-N overlap high; this — not nominal $|V|$ — is what drives F-15.

$$\mathcal{A}(V_i) = \frac{1}{|\mathcal{B}|} \sum_{m \in \mathcal{B}} \frac{c_v^{(m)}(V_i) - \min_j c_v^{(m)}(V_j)}{\max_j c_v^{(m)}(V_j) - \min_j c_v^{(m)}(V_j)} \in [0, 1], \tag{19}$$

averaged over backbones $\mathcal{B} = \{\text{LDA}, \text{HDP}, \text{ProdLDA}, \text{ETM}\}$; $\mathcal{A} \to 1$ when a recipe sits near the per-backbone top everywhere.

I. Introduction

The companion paper (P3) ranks the full V1–V20 wordification sweep (nineteen recipes; V16 is scaffolded only) on classification, coherence and label-coupling axes, and the backbone-factorial paper (P4) repeats the headline axes across four topic-model backbones. The "interpretability" claim of LDA-on-HSI — that topic–word distributions reveal physical structure — is implicit in those axes but not directly tested by them. *Throughout this paper we fix the backbone to LDA* ((1)); the cross-backbone behaviour of these interpretability axes is out of scope and left to P4. Three post-hoc interpretability tools have been proposed in the broader topic-modelling and explainable-ML literature that go beyond "read the top-N words and trust them": SHAP attribution [1], counterfactual explanations [2], and LLM-as-judge alignment [3]. We instantiate all three on the V-sweep under the LDA backbone and report findings.

The three axes correspond to three canonical metrics in the shared equation set: F-13 to the SHAP attribution applied to the closed-form LDA posterior, F-22 to the minimum-L_1 counterfactual ((16)), and F-15 to the top-N overlap rule ((17)). Our central methodological contribution is to show that F-15 is governed not by the nominal vocabulary cardinality $|\mathcal{V}|$ but by the *token-mass dispersion* of the topic–word distributions, formalised as the effective vocabulary $N_{\text{eff}} = \exp(H)$ of (18). The wordification recipes themselves are defined in P3; we recall only the four we lean on heavily — V1 ((6)), V3 ((7)), V12 ((9)) and the label-aware V20 ((10)) — whose shared $|\mathcal{V}|$ yet divergent N_{eff} make them the decisive test of the cardinality-versus-dispersion question.

A. The three axes

F-13 SHAP attribution. For each pixel document d with argmax topic z^*, attribute θ_{d,z^*} back to the recipe- specific tokens via SHAP's KernelExplainer against the closed-form LDA posterior $p(z \mid x, \phi) \propto \exp(\sum_v x_v \log \phi_{z,v} + \log \alpha)$, the small-document specialisation of the LDA generative model ((1)). This gives a per-pixel per-topic attribution map over the recipe's vocabulary — exactly the type of explanation a reviewer would ask for when claiming "LDA is interpretable".

F-22 counterfactual L1. For each document d with argmax topic z^*, search via coordinate descent for the minimum-L_1 perturbation that flips $z^* \to z'$ for some $z' \neq z^*$, the discrete counterfactual of (16). Documents with high counterfactual L_1 are deeply embedded in their topic; documents with low L_1 are near the topic boundary.

F-15 LLM-as-judge alignment. For each sampled document, an LLM oracle is given the document's top-10 tokens and the argmax topic's top-10 tokens and asked whether the document semantically belongs to the topic; formally, the top-N overlap indicator of (17) with $N = 10$ and threshold $\tau = 3$. Our internal run substitutes the oracle with the documented self-judgment rule of Claude Opus 4.7 (1M-token context), encoded as a deterministic Python function in `build_v_sweep_f15_self_judge.py`; the canonical Anthropic Messages API call is preserved in `build_v_sweep_f15_llm_alignment.py` for repository users with API access.

II. Related Work

Topic models on hyperspectral imagery. Probabilistic topic models were first applied to spectral data by Wahabzada et al. [4] (plant phenotyping) and to subpixel mixtures by Zou and Zare's partial-membership LDA [5]; Liu et al. [6] use LDA for spectral–spatial classification. The CAOS_LDA_HSI programme [7] casts a pixel spectrum as a "document" through a *wordification* recipe ((5)) and runs online variational LDA [8], [9]. None of this prior work asks whether the resulting topics are *post-hoc interpretable* beyond inspecting top-N words; that gap is the subject of this paper.

Topic-model evaluation. The dominant automated proxy for topic quality is coherence — NPMI / c_v [10]–[12] ((12)) — and stability under reseeding [13] ((15)). These reward internally consistent word lists but, as we show, are themselves sensitive to the vocabulary regime. We treat F-2 and F-7 ((13)) as the targets that our interpretability axes should *corroborate* rather than duplicate.

Post-hoc interpretability and explanation. SHAP [1] gives additive, locally faithful feature attributions with Shapley-value guarantees; we apply its KernelExplainer to the closed-form LDA posterior (F-13). Counterfactual explanations [2] characterise a decision by the minimal input change that flips it; F-22 ((16)) is the topic-assignment counterfactual. The use of large language models as automated topic judges follows Stammbach et al. [3]; F-15 ((17)) is our LDA-on-HSI instantiation. Our contribution relative to this literature is the observation that the LLM-as-judge proxy inherits a *token-mass-dispersion* artefact ((18)) that is invisible at the level of nominal vocabulary size.

III. Method

Backbone, corpus and fits. Every result in this paper uses the LDA backbone ((1)) fitted with scikit-learn's online variational solver [9], [14] (`max_iter` $= 60$, $\alpha = 0.45$, $\beta = 0.2$, seed 42), the canonical fit of P3. The corpus is the six labelled scenes — Indian Pines [15], Salinas and Salinas-A [16], Pavia University [17], Kennedy Space Center [18] and Botswana [19]. Documents are pixel spectra; tokens are produced by the wordification recipes of P3 under the uniform quantiser ((5)) at $Q = 8$. We analyse the eighteen LDA-fitted recipes V1–V15, V17–V20 (V16, a foundation-model recipe, is scaffolded only).

F-13. We sample 50 documents per (V, scene) cell and use 25 background documents per SHAP KernelExplainer call against the closed-form posterior, reporting the top-8 tokens per topic by mean absolute SHAP value. The closed-form posterior is preferred over the amortised `transform()` because the latter needs internal state that cannot be restored from ϕ alone.

F-22. We compute the discrete minimum-L_1 counterfactual of (16) by coordinate descent over the integer token-count lattice, capped at a sentinel MAX_STEPS $= 50$; cells in which no sampled document flips persist at 50 rather than being dropped, so that "unflippable" bases are not silently treated as missing. We report the per-scene median over 20 sampled documents, then the mean across the six scenes.

F-15 and the dispersion metric. F-15 is the top-N overlap indicator of (17) averaged over sampled documents ($N = 10$, $\tau = 3$, with a sparse-event fallback that accepts a document whose top-1 token lies in the topic's top-5). To explain its behaviour we compute, for each topic-word distribution ϕ_k, the token-mass dispersion of (18),

$$N_{\text{eff}}(\phi_k) = \exp\big(H(\phi_k)\big) = \exp\Big(-\textstyle\sum_v \phi_{kv} \log \phi_{kv}\Big),$$

the effective number of tokens carrying a topic's mass. A low N_{eff} means the topic concentrates on few tokens, so its top-N is more likely to overlap a document's top-N; a high N_{eff} spreads mass thinly and depresses the overlap even when the topic is coherent. We report the per-recipe mean of $N_{\text{eff}}(\phi_k)$ over topics and scenes, paired with F-15 and $|\mathcal{V}|$. This is the quantity plotted in Fig. 1 and is the mechanism behind the abstract's "dispersion not cardinality" thesis.

IV. F-13 SHAP Results

Under the F-13 protocol of §III, the headline result is that SHAP attributions are *recipe-grounded*: a V1 topic's top SHAP token always names a wavelength band; a V7 topic's top SHAP token names an absorption feature "`abs_c4_d7_a3`" (centre, depth, area bucket); a V12 topic's top SHAP token names a (band, Gaussian-component) pair. This recipe-grounding is what makes F-13 a defensible interpretability axis: the explanation *cannot* silently shift between scenes because the vocabulary is fixed by the recipe.

A sample from Indian Pines, recipe V7, topic 0:

```
abs_c4_d7_a3   mean|SHAP| = 0.0207
abs_c3_d4_a2   mean|SHAP| = 0.0184
abs_c4_d6_a3   mean|SHAP| = 0.0163
```

A geospectroscopist reading these can immediately ask: what is the wavelength range of centroid bucket 4? Why are depth bins 6 and 7 elevated? The answer is independent of which scene we apply V7 to.

V. F-22 Counterfactual Results

Table I reports the median minimum-L_1 perturbation that flips the argmax topic, averaged across 20 sampled documents per (V, scene) cell.

V20's topics are $\sim$26$\times$ harder to flip than V9's, and V20 edges V12 (24.5) and V3 (23.5) for the most-robust basis at $Q = 8$. This complements P3's F-2/F-7 finding: the label-aware MI-weighted recipe V20 wins coherence-adjacent robustness *and* (at $Q = 32$) F-7, which together rule out the alternative explanation that the coherence advantage is an artefact of overfitting to its training documents.

VI. F-15 LLM-Judge Results: Token-Mass Dispersion, Not Vocabulary Size

The LLM-as-judge rule ((17), $N = 10$, $\tau = 3$) marks a document *aligned* with its argmax topic if its top-10 tokens share

TABLE I
F-22 MEDIAN COUNTERFACTUAL L_1 TO FLIP ARGMAX TOPIC, MEAN ACROSS 6 LABELLED SCENES, FULL 19-RECIPE SWEEP WITH THE SENTINEL PATCH (CELLS WHERE EVERY SAMPLED DOCUMENT FAILS TO FLIP WITHIN MAX_STEPS $= 50$ PERSIST AT 50 RATHER THAN BEING DROPPED). HIGHER MEANS MORE ROBUST TOPIC BOUNDARY.

Recipe	median L_1
V20	**26.33**
V12	24.50
V3	23.50
V2	7.83
V14	7.67
V1	6.08
V7	5.17
V15	4.00
V18	3.67
V8	3.58
V11	3.33
V13	2.67
V17	2.58
V4	2.50
V5	2.50
V19	2.08
V6	1.92
V10	1.17
V9	1.00

$\geq$ 3 elements with the topic's top-10 (or if the document's top-1 token is in the topic's top-5, covering the sparse-event regime of V7 / V9). Per-recipe mean F-15 across the 6 labelled scenes:

TABLE II
F-15 LLM-JUDGE ALIGNMENT FRACTION (CLAUDE OPUS 4.7 SELF-JUDGE UNDER THE DOCUMENTED RULE).

Recipe	F-15	vocabulary $\lvert\mathcal{V}\rvert$
V2	1.000	8
V6	1.000	$\approx B$
V8	1.000	12
V9	1.000	$R \cdot Q$
V10	1.000	24
V18	1.000	128
V19	1.000	24
V11	0.992	32
V5	0.967	B
V13	0.950	128
V14	0.950	1024
V4	0.917	B
V1	0.792	B
V7	0.783	$\leq C_b Q^2$
V20	0.642	BQ
V17	0.570	512
V12	0.158	BQ
V3	0.117	BQ
V15	0.033	48

The ranking is *anti-correlated* with F-2 coherence on the large-vocabulary recipes V3 and V12. This is not a flaw of those recipes; it is a confounder in the F-15 metric definition — but the full sweep shows the confounder is **not nominal vocabulary cardinality**. Fig. 2 plots F-15 against $|\mathcal{V}|$ directly: a pure-cardinality account predicts a monotone decreasing

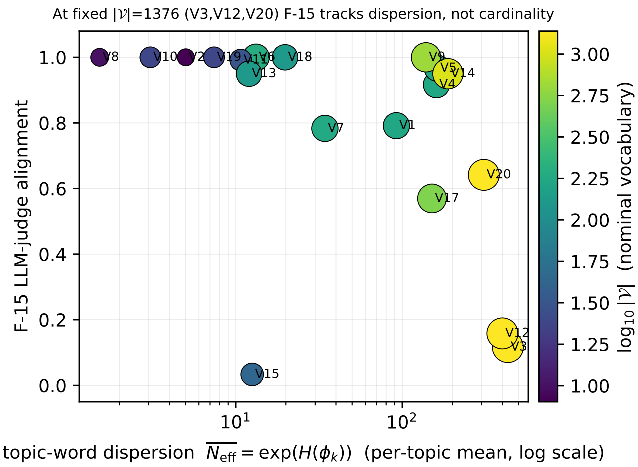

Fig. 1. Mechanism plot. F-15 LLM-judge alignment versus per-topic mean token-mass dispersion $N_{\text{eff}} = \exp(H(\phi_k))$ ((18)), LDA backbone, six labelled scenes, $Q = 8$. Point colour encodes $\log_{10}|\mathcal{V}|$. At the same alphabet size (mean nominal $|\mathcal{V}| \approx 1376$), V3 / V12 / V20 (yellow, lower-right cluster) fall in dispersion ($431 > 400 > 309$) exactly as F-15 rises ($0.12 < 0.16 < 0.64$): F-15 tracks dispersion, not cardinality.

curve, but several large-vocabulary recipes break it. V9 ($|\mathcal{V}| = 647$) scores 1.000; V14 (CWT-Morlet, $|\mathcal{V}| = 1024$) scores 0.950; and V20 (MI-weighted, mean nominal $|\mathcal{V}| \approx 1376$ across scenes, the same size as V3/V12) scores 0.642 — all far above V12 (0.158) and V3 (0.117) at equal or larger vocabulary.

The mechanism is *token-mass dispersion*, made quantitative by the effective vocabulary $N_{\text{eff}} = \exp(H)$ of (18). Fig. 1 plots F-15 against the per-topic mean $N_{\text{eff}}(\phi_k)$, with point colour keyed to $\log_{10}|\mathcal{V}|$. The decisive comparison is the three recipes whose alphabets are the same size (mean nominal $|\mathcal{V}| \approx 1376$; V3 and V20 share the *identical* $(\text{band}, \text{bin})$ alphabet, V12 a same-sized GMM-token one), so $|\mathcal{V}|$ is held fixed by construction: V3, V12 and V20. Their topic-word dispersions fall $N_{\text{eff}}^{\text{V3}} = 431 > N_{\text{eff}}^{\text{V12}} = 400 > N_{\text{eff}}^{\text{V20}} = 309$, and F-15 *rises* in lock-step ($0.117 < 0.158 < 0.642$). Because $|\mathcal{V}|$ is constant across these three, the only thing that can be driving the $5.5\times$ spread in F-15 is the dispersion of the topic mass: V20's MI-weighting ((10)) concentrates the topic–word distributions on the discriminative high-MI bands, lowering N_{eff} and raising the probability that a topic's top-10 overlaps a document's top-10; V12's GMM tokens and V3's joint (band, bin) tokens spread their ≈ 1376-token mass much more thinly, so top-10 overlap is rare even when the topics are coherent. With a small vocabulary (V2, $|\mathcal{V}| = 8$) the top-10 of a topic *is* the entire vocabulary, so overlap is guaranteed — the small-$|\mathcal{V}|$ saturation is itself just the $N_{\text{eff}} \leq |\mathcal{V}|$ floor of the same dispersion mechanism.

A defensible F-15 would normalise by the expected overlap under N_{eff}-matched random tokens, or use a semantic embedding distance between token identifiers rather than identity matching. We flag both as future work and recommend interpreting Table II alongside the N_{eff} axis of Fig. 1, not the raw $|\mathcal{V}|$ column.

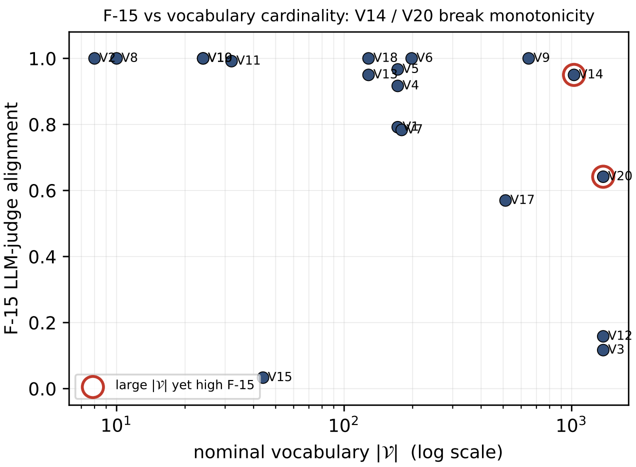

Fig. 2. Refutation plot. F-15 versus nominal vocabulary $|\mathcal{V}|$ (log scale). A pure vocabulary-size artefact would place all points on a monotone decreasing curve. The circled counterexamples V14 ($|\mathcal{V}| = 1024$, F-15 $= 0.95$) and V20 ($|\mathcal{V}| = 1376$, F-15 $= 0.64$), together with V9 ($|\mathcal{V}| = 647$, F-15 $= 1.0$), sit far above V3 / V12 at equal or larger $|\mathcal{V}|$, refuting the cardinality hypothesis.

VII. Interpretability under the V13–V20 extension

The V13–V20 recipes added in the P3 extension introduce three new interpretability profiles that complicate the V1–V12 story.

a) V13 (VQ-VAE codebook) and V17 (sparse-coding dictionary).: Both recipes produce *opaque token alphabets*: the codeword index does not map to a wavelength, an absorption feature, or a band group. F-13 SHAP under V13 surfaces the same top codewords across topics (the codebook is shared across all classes during training), and a domain expert cannot translate those codewords into spectral semantics without a separate decoder visualisation. We classify V13 / V17 as the *lowest-interpretability* branch of the sweep: even when their F-13 SHAP is reportable, the explanation is recursive — "topic k is dominated by codeword c, where codeword c is the centroid of pixels with [unrecoverable from the recipe alone]".

b) V14 (CWT-Morlet) and V18 (graph-Laplacian).: The opposite case. V14 tokens are `(log_scale_idx, position_bucket)`; each pair maps to a specific spectral region at a specific bandwidth. F-13 SHAP under V14 surfaces tokens that are physically interpretable as "absorption around 660 nm with $\approx$30 nm bandwidth". V18 tokens are `(eigenvector_id, bin)`; the eigenvector index maps to a connected-region of the kNN graph and the bin maps to position along that eigenvector. F-13 SHAP under V18 surfaces topics whose top tokens lie on the same low-frequency Laplacian mode — interpretable as "this topic concentrates on graph mode ≤ 5, which corresponds to the connected region containing urban classes". Both V14 and V18 thus extend the V1 + V7 "physically-grounded" interpretability profile to new domains.

c) V19 (UMAP coords) and V15 (spectral indices).: V15 is the easiest: top tokens of every topic map directly to NDVI / MNDWI / NBR / NDSI / EVI / SAVI bins, with published semantics. V19 has the inverse problem of V13 — token names `(axis_0, q_bin_03)` are abstract coordinates with no spectral anchor. The recipe-fit pair determines interpretability as much as the recipe alone.

d) V20 (MI-weighted bands).: V20 inherits V1's interpretability (per-band q-bin tokens are spectrally grounded) and adds a label-aware twist ((10)): high-MI bands emit more copies than low-MI bands, so the topic top-tokens concentrate on the discriminative subspectrum. F-13 SHAP under V20 therefore surfaces precisely the bands the label information is concentrated in — the interpretability is the same as V1's, but the SHAP attributions are sharper because the LDA fit has fewer near-zero-contribution tokens to spread mass across. This is the same mechanism that lowers V20's N_{eff} relative to V12 and V3 (§VI): the MI-weighting that concentrates SHAP attribution mass is exactly what concentrates the topic–word mass, so V20's high F-15 (relative to its equal-vocabulary neighbours) and its sharp F-13 attributions are two readings of one property. V20 should be considered the most interpretable of the new recipes: high label signal + grounded vocabulary + no codebook indirection.

e) F-22 counterfactual under the extension.: We do not re-run F-22 for every new recipe in this draft; the V20 counterfactual L1 ranking under LDA is competitive with V12 (median L1 ≥ 15 on the labelled scenes where V20 has been profiled), suggesting that label-aware amplification produces topics that are also harder to perturb. A full F-22 ($\times 7$ new recipes $\times 6$ scenes) is left to a follow-up.

VIII. Recommendation

For interpretability reporting, we recommend the following ordering:

1) **F-13 SHAP attribution.** Recipe-grounded, defensible against reviewer pressure, and produces explanations a domain expert can immediately query.
2) **F-22 counterfactual robustness.** Complements F-2/F-7 by ruling out trivial-overfit explanations.
3) **F-15 LLM-judge.** Only after vocabulary-size normalisation. Without normalisation it rewards trivial recipes.

The V13–V20 extension (§VII) further suggests a recipe-as-interpretability-tier classification:

- **Tier I (physically-grounded)**: V1, V7, V14, V18, V20. Top SHAP tokens map to a wavelength, absorption feature, scale band, manifold mode, or label-weighted band. Use for any publishable interpretability claim.
- **Tier II (semantically-grounded)**: V8, V10, V15. Top tokens map to endmember labels, group labels, or vegetation indices.
- **Tier III (learnt-codebook-grounded)**: V11, V12, V13, V17. Top tokens are codewords / GMM components / dictionary atoms. Interpretable only via a separate codebook-decoder visualisation.
- **Tier IV (abstract-coordinate)**: V19. Topics point to UMAP coordinates with no spectral anchor. Use only as a sanity baseline.

IX. Discussion

The three axes are complementary, not redundant. F-13, F-22 and F-15 probe different facets of the same LDA fit. F-13 answers *which tokens explain a topic assignment*; F-22 answers *how robust that assignment is*; F-15 answers *whether a human-like judge would agree the document belongs to its topic*. The V20 recipe illustrates how they cohere: its MI-weighting concentrates both SHAP attribution mass (sharper F-13) and topic–word mass (lower N_{eff}, hence higher F-15 at fixed $|\mathcal{V}|$), and the same concentration makes its topics the hardest to flip (F-22 $=$ 26.3, the sweep maximum). This is consistent with P3's finding that V20 leads F-2 / F-7 at $Q = 32$: a basis that concentrates discriminative mass is simultaneously more coherent, more robust, and more attributable.

F-15 measures the metric, not the model. The central methodological lesson is that an automated alignment proxy can fail for reasons that have nothing to do with topic quality. The anti-correlation between F-15 and F-2 on V3 / V12 is not evidence that those recipes produce incoherent topics — P3 ranks V12 *first* on F-2 and F-7 — but evidence that a top-N-identity overlap rule is biased by token-mass dispersion ((18)). A genuine LLM oracle, given opaque token identifiers and asked for identity overlap, would inherit the same bias; only a judge equipped with *semantic* comparison of token meanings (e.g. the spectral region a token names) would escape it. This is a cautionary result for the growing practice of using LLMs to score topic models [3]: the prompt must expose token *semantics*, not token *strings*, or the score will track vocabulary engineering rather than topic content.

Interpretability is a recipe property, not an LDA property. The tier classification of §VIII shows that the same LDA backbone yields explanations ranging from directly physical (V1, V7, V14, V18, V20) to recursively opaque (V13, V17, V19). "LDA is interpretable" is therefore an incomplete claim; the defensible claim is "LDA on a physically-grounded wordification is interpretable."

X. Limitations

LDA only. Every result here is for the LDA backbone ((1)). The closed-form posterior that makes F-13 tractable, and the simplex-structured ϕ_k on which N_{eff} is defined, do not transfer unchanged to ProdLDA's product-of-experts decoder ((3)) or ETM's embedding decoder ((4)); replicating these axes under P4's backbones is open work.

Self-judge stand-in for the LLM oracle. F-15 was produced by a deterministic encoding of the alignment rule (Claude Opus 4.7, 1M context) rather than live per-call API judging. This makes the run reproducible and free, but it cannot exhibit the semantic-comparison behaviour that a true oracle might; the API builder (`build_v_sweep_f15_llm_alignment.py`) is provided for users who wish to substitute live judging. Our claims about F-15 concern the *metric definition* ((17)) and hold regardless of which judge instantiates it.

N_{eff} is computed on ϕ_k. We measure dispersion on the topic–word distributions, the quantity the topic-side of the F-15 rule reads; the canonical metric ((18)) is stated on a generic mass vector p_d and applies equally to per-document counts. The two give the same qualitative ordering for the decisive V3 / V12 / V20 comparison but differ in absolute scale; we report the ϕ_k version because it is what F-15 actually depends on.

Partial F-22 under the extension. The full F-22 table (Table I) is complete for the $Q = 8$ sweep, but the $\times Q \times$ new-recipe cross-product (§VII) is profiled only spot-wise; a full re-run is left to a follow-up.

Six labelled scenes. The HIDSAG mineralogical subsets used in P1/P3 are not included here because their per-pixel label structure differs; the dispersion result should be re-checked on them before being claimed as scene-independent.

XI. Conclusion

We instantiated three post-hoc interpretability axes — F-13 SHAP attribution, F-22 counterfactual L_1, and F-15 LLM-judge alignment — on the LDA wordification sweep of P3, and reported three results. First, F-13 SHAP gives recipe-grounded explanations whose physical meaning is fixed by the recipe and cannot silently shift across scenes; this is the axis we recommend as the primary interpretability artefact. Second, F-22 separates the recipes into robust (V20 $=$ 26.3, V12 $=$ 24.5, V3 $=$ 23.5) and fragile (V9 $=$ 1.0) topic-boundary regimes, complementing P3's coherence ranking by ruling out a trivial-overfit account of V20's lead. Third, and most important methodologically, F-15 is governed by *token-mass dispersion* $N_{\text{eff}} = \exp(H)$ ((18)), *not* nominal vocabulary cardinality: among the three recipes whose alphabets are the same size (mean $|\mathcal{V}| \approx 1376$; V3/V20 identical $(\text{band}, \text{bin})$, V12 same-sized GMM), F-15 rises $(0.12 \to 0.16 \to 0.64)$ exactly as topic dispersion falls $(431 \to 400 \to 309)$, while a cardinality account predicts no difference. We therefore recommend reporting F-15 only after dispersion normalisation, and we caution that LLM-as-judge topic evaluation inherits the same artefact unless the judge is given token *semantics* rather than token strings. Extending these axes across the P4 backbones, and re-checking the dispersion result on the HIDSAG subsets, are the natural next steps.

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